

ARCHITECT REVIEW

SALDEV-1499 | Quote Line Type Flow Deactivation

Flywire Partial sandbox | Read-only implementation and promotion-readiness assessment

Environment	FlywirePartial Org 00DhG0000000jOX Sandbox
Review date	28 August 2026
Evidence	Live FlowDefinition/Flow/Apex read-back unit-test results coverage source inspection
Decision	Changes Requested - not ready for promotion
Boundary	No code edits, deployment, activation, data changes, or Production changes

ARCHITECT DECISION

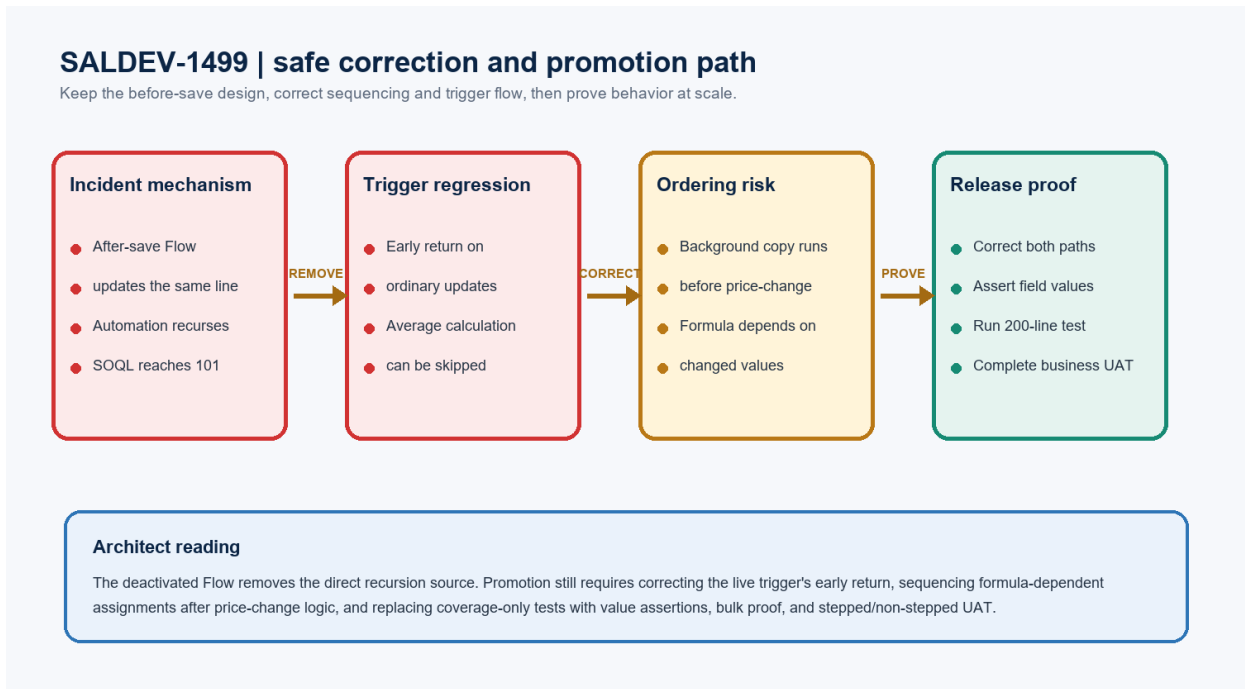
Do not promote yet. The legacy Flow is genuinely inactive and the before-save Apex direction removes its extra DML cycle, but the deployed trigger can skip existing rollup logic, formula-dependent fields are copied in the wrong sequence, and the tests do not prove the required field outcomes.

Acceptance outcome

#	Scope	Result	Evidence
1	Flow deactivation	Pass	FlowDefinition.ActiveVersionId is null; latest v13 is Obsolete.
2	Root-cause direction	Pass	Background fields moved to before-save memory assignment; no Flow self-update remains.
3	Trigger regression safety	Fail	An after-update early return can bypass calculateAveragePrice().
4	Functional sequencing	Fail	Quote Line Type is copied before price-change logic updates a field used by its formula.
5	Automated proof	Partial	Eight tests pass, but SALDEV-1499 assertions do not verify the output fields or 200-line limits.

Incident mechanism and control path

The incident was raised after an after-save record-triggered Flow updated the same Quote Line that initiated the transaction. That second DML operation re-entered Quote Line automation and the transaction exhausted the 100-query synchronous limit. Moving simple assignments to a before-save trigger is the correct architectural direction because it modifies Trigger.new without a second update.



Current SALDEV-1499 implementation evidence

Flow and Apex state

Evidence area	Observed state	Result	Architect reading
Target org	FlywirePartial; sandbox; USA1148S	Pass	Verified against Organization Id 00DhG0000000jOXUAY.
Flow definition	ActiveVersionId = null	Pass	The Flow is genuinely deactivated.
Latest Flow version	v13; API 65; Obsolete	Pass	No active version remains.
QuoteLineTrigger	Active; API 67	Partial	Before-save replacement exists, but after-update control flow can exit early.
Trigger handler	Active; API 67	Partial	Bulk collection exists; background-field order and behavior require correction.
Failed Quote Line	a1zhG000000MBi7QAG not found	Partial	The exact incident record cannot be replayed or field-read back.

Automated test and coverage evidence

Evidence area	Observed state	Result	Architect reading
Latest focused run	8 test methods passed	Pass	Latest results recorded 27 Aug 2026 around 21:15 UTC.
Trigger coverage	23 covered / 10 uncovered	Partial	69.7% aggregate coverage; uncovered branches include control paths.
Handler coverage	245 covered / 24 uncovered	Pass	91.1% aggregate coverage, but percentage does not prove behavior.
SALDEV-1499 assertions	Record existence / non-empty lists	Fail	Expected background, mismatch, fallback, and price-change values are not asserted.
Bulk governor proof	No 200-line limits test	Fail	The original SOQL-101 acceptance concern is not directly proven.

EVIDENCE BOUNDARY

The linked design document is not accessible to the connected Google account. Requirement alignment for fallback and adjusted-rate behavior therefore remains conditional until the team shares the document or records the decisions in Jira.

Detailed code-review findings

Priority	Finding	Why it matters	Required correction
P1	After-update early return	When OLI synchronization is unnecessary, the trigger returns before calculateAveragePrice(). Ordinary updates can stop recalculating existing rollups.	Replace the trigger-level return with a guarded OLI-sync block so later after-save logic always reaches its own recursion guard.
P1	Formula copy occurs too early	processBackgroundFields() copies Quote_Line_Type__c before handlePriceChangeOnAmendmentClones() updates Price_Change__c. The live formula depends on Price_Change__c and Group_Allow_Product_Ramping__c.	Compute price-change inputs first, then derive/copy Quote Line Type from final values, or calculate the classification explicitly in one method.
P1	Tests do not assert outcomes	Passing tests can still leave Quote_Line_Type_background__c, Rate_Details_Changed__c, or Default_Rate_Details__c wrong.	Re-query records and assert exact true/false/text values for insert, update, ramp, renewal, amendment, cancellation, and blank cases.
P2	Behavior exceeds the legacy Flow	Apex adds default-rate fallback and adjusted-rate override comparisons beyond the Flow's direct Rate_Details versus Default_Rate_Details check.	Confirm these rules in the design/Jira acceptance criteria and test each branch; otherwise narrow the implementation.
P2	Local source is stale	The checkout still shows Flow v13 Active and an older after-only trigger. It does not match the live sandbox implementation.	Retrieve the exact live components into a focused review package before producing the promotion artifact or diff.

PROMOTION BLOCKER

Do not approve promotion until the trigger return and formula-ordering defects are corrected, explicit field-value assertions pass, and a bulk transaction demonstrates that the replacement stays below governor limits.

Functional equivalence and implementation boundary

Legacy-to-Apex behavior that must remain equivalent

Behavior	Legacy Flow	Replacement expectation
Quote Line Type background	Copy formula value after save when blank or different	Persist the classification calculated from the transaction's final price-change/ramping state.
Rate mismatch	Rate_Details__c != Default_Rate_Details__c	Match approved null/blank semantics and document any fallback/override expansion.
Record update	Explicit DML on the initiating record	Before-save memory assignment only; no self-DML.
Existing rollups	Separate trigger behavior	Continue calculateAveragePrice() on all intended insert/update/delete/undelete paths.

Focused correction scope

Component type	Focused scope	Required action
Quote Line trigger	Before-save call order	Run price-change classification before the background copy, or combine both deterministically.
Quote Line trigger	After-update OLI-sync branch	Remove the trigger-level early return and preserve subsequent rollup execution.
Trigger handler	processBackgroundFields	Define null/blank, fallback, adjusted-rate, and formula-copy behavior explicitly.
Apex tests	QuoteLineTriggerHandleTest	Assert exact outputs and prior rollup behavior; avoid pass-by-record-existence assertions.
Bulk proof	200 mixed Quote Lines	Capture query/DML counts and verify all classifications, mismatch flags, and rollups.
Focused package	Trigger, handler, test, inactive Flow state	Retrieve current live metadata and validate only the reviewed components plus required dependencies.

Release sequence and production gate

Phase	Required evidence
Requirement confirmation	Share the design document or copy its approved decisions into Jira: formula timing, blank handling, fallback source, adjusted-rate rules, and stepped-up behavior.
Focused correction	Surgically correct trigger order/control flow and tests; keep the Flow deactivated and preserve unrelated CPQ automation.
Sandbox validation	Run the focused Apex class plus relevant Quote Line regression tests; record test-run ID, coverage, query/DML headroom, and field-level results.
Business UAT	Create stepped and non-stepped quotes; save/recalculate; confirm no SOQL-101 error and verify Quote Line Type, mismatch flags, and existing averages.
Promotion/read-back	After approval only: validate the production-target package, deploy, verify Flow ActiveVersionId remains null, read back Apex versions, and repeat smoke tests.

Validation evidence to capture

Evidence	Required detail
Automated test receipt	Test-run ID, named methods, pass/fail totals, runtime, and exact trigger/handler coverage.
Governor headroom	Queries and DML statements before/after the 200-line transaction, with no SOQL-101 exception.
Record-level read-back	Quote and Quote Line IDs plus expected/actual background type, mismatch flag, default rate details, and average fields.
Metadata state	FlowDefinition.ActiveVersionId, latest Flow status, Apex API/status/modified date, and reviewed package component count.

PRODUCTION GATE

This review covered FlywirePartial only. Production was not inventoried or changed. Re-retrieve current Production metadata and run focused check-only validation before any deployment decision.

UAT and approval gates

Test	Pass evidence
Ordinary Quote Line update	An update with no OLI-sync requirement still runs calculateAveragePrice() and preserves expected average placement.
Stepped-up renewal	A ramp-enabled renewal saves without error and background type equals the final Quote_Line_Type__c classification.
Amendment and cancellation	Price Change, Cancellation, and Cancellation Price Change outcomes match final quantity and price-change inputs.
Rate details equal/different	Equal values set Rate_Details_Changed__c false; approved mismatch/fallback/override cases set it true.
Blank and null handling	Blank formula/default/source/subscription cases produce documented values without stale background data or exceptions.
Bulk and limits	200 mixed Quote Lines complete in one transaction with correct fields, rollups, query headroom, and no SOQL-101 error.
Non-stepped regression	Existing ordinary quoting and average-price behavior match the approved sandbox baseline.

Required approval gates

#	Required action	Owner	Evidence needed
1	Confirm expanded rate-detail rules	Product Owner / Architect	Accessible design decision or Jira acceptance detail.
2	Correct trigger control flow/order	Development Lead	Reviewed focused diff with no unrelated rewrite.
3	Strengthen automated tests	Developer / QA	Exact field assertions, regression results, and 200-line limits evidence.
4	Complete business UAT	CPQ Business Owner	Quote IDs, scenarios, expected/actual values, and no-error confirmation.
5	Approve promotion	Architect / Release Manager	Check-only job ID, component/test counts, exclusions, and sign-offs.

Ready-to-paste architect response

STORY COMMENT

Architect review completed for SALDEV-1499 in FlywirePartial. Changes Requested. The Quote Line Type background Flow is correctly deactivated (ActiveVersionId is null; v13 is Obsolete), and the before-save Apex direction removes the Flow's self-update. However, the deployed after-update trigger can return before calculateAveragePrice(), and processBackgroundFields() copies the formula-backed Quote Line Type before price-change logic updates fields used by that formula. Eight focused tests pass, but the SALDEV-1499 tests do not assert the resulting background/rate-detail fields and there is no 200-line governor-limit proof. Correct the trigger control flow and sequencing, add exact field-value and bulk tests, share the detailed design decision, and complete stepped/non-stepped UAT before promotion. No code, metadata, activation, deployment, data, or Production changes were made during this review.